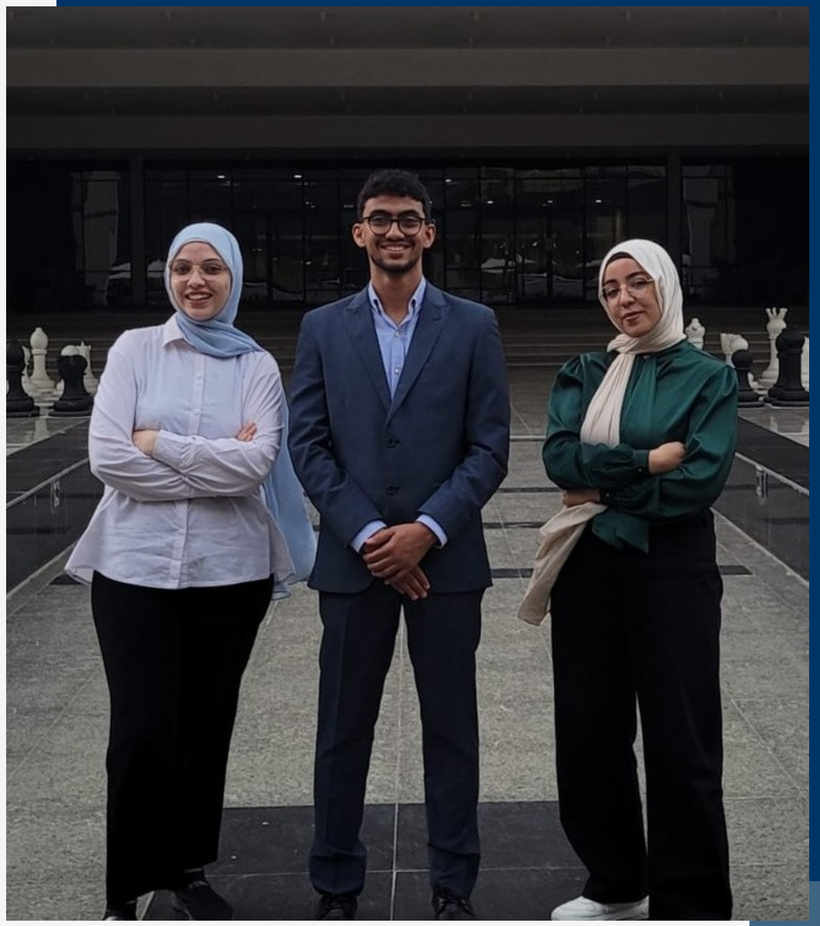
مدينة زويل
ZEWAIL CITY

EUROAVIA
CAIRO

EUROPEAN
ROTORS
THE VTOL
SHOW AND
SAFETY
CONFERENCE

THE AVALANCHE



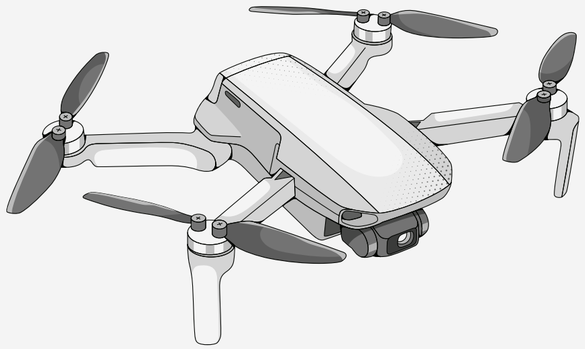


TABLE OF CONTENTS

Research Findings

Project Design

Public Engagement Strategies

ABOUT US

Team members

- Yahya Hazem
- Eman Yasser
- Asmaa Sapry

Country

- Egypt

University

- Bsc Aerospace Engineering
- University of Science and Technology in Zewail City

• VISION

Partaking in the ROTORTHON challenge, we do not only look forward to enhance our technical skills in the aerospace field, but also improve our problem-solving skills. Through our involvement, our goal is to enhance rotorcraft technology and raise Egypt's position in the international aerospace industry.

• MISSION

Our team is committed to advancing innovation in unmanned systems while supporting the United Nations Sustainable Development Goals, with a focus on promoting gender equality and increasing women's participation in STEM within aerospace. We aim to address critical safety and privacy issues in the UAV industry, tackle challenges in U-Space integration, and develop solutions that enhance public acceptance, reduce noise, and improve safety. Through our work, we strive to combine sustainability, social responsibility, and technological progress in shaping the future of aerospace.

FACTORS INFLUENCING PUBLIC ACCEPTANCE OF UAVS

Drones have become increasingly commonplace among the general population, facilitating enhanced connectivity and intelligence within urban environments. Yet, we seem to face some challenges regarding:

- **Safety protocols**

Public concerns about drone safety arise from the risk of crashes and collisions that may result in bodily harm and property damage. These issues highlight the need for strict regulations and public education to address safety fears.

- **Privacy concerns**

People often oppose drones due to privacy concerns, fearing they may be spied on or recorded without consent. This belief fuels discomfort and resistance to UAVs. Additionally, drones are unwelcome in certain areas, such as military bases, where they may be shot down for national security.

- **Noise**

Different species react differently to various noise levels and frequencies. Prolonged exposure to sounds above 85 decibels can damage human hearing. While humans hear between 20 and 20,000 Hz, birds can detect sounds beyond human hearing, even in the ultrasonic range. Dogs have sensitive hearing, able to perceive frequencies from 40 Hz to 60,000 Hz, and may feel stressed at noise levels between 60 and 80 dB. Understanding these sensitivities is crucial for protecting animal health and reducing noise pollution's impact.

- **Environmental Impact**

EXISTING U-SPACE PROJECTS

- **Convo 5**

The Convo 5 Project highlights the potential of U-Space principles to transform urban air mobility. By focusing on safety, efficiency, and community engagement, this initiative has set a precedent for integrating UAVs into urban environments, paving the way for a future where drones coexist harmoniously with traditional air traffic.



- **AURA**

The AURA (PJ.34) project focuses on integrating the management of very low-level airspace (VLL), also referred to as U-space, with conventional air traffic management (ATM) systems. Its aim is to facilitate smooth operations between U-space and ATM.

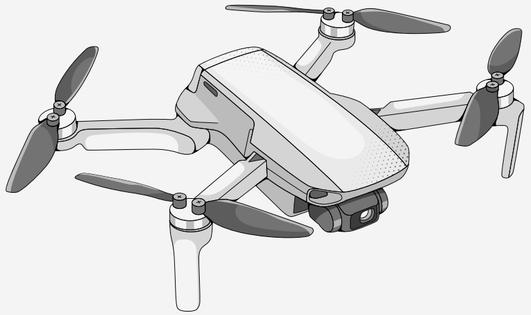
IMPLEMENTING U-SPACE IN A “HYPOTHETICAL CITY”

- **Objectives**

1. **Safety Assurance**: Establish robust protocols and systems to ensure the safety of both drone operations and the public.
2. **Privacy Protection**: Implement measures to safeguard personal privacy and prevent unauthorized surveillance.
3. **Noise Mitigation**: Develop strategies to reduce the noise generated by UAVs, especially in densely populated areas.
4. **Environmental Management**: Minimize the ecological footprint of drone operations and address potential impacts on local wildlife.
5. **Regulatory Framework**: Create adaptive regulations that support innovation while ensuring public safety and privacy.


Storyblocks

ENHANCED SAFETY MEASURES



- **Identification & Tracking**

Each UAV should have a unique identification number, similar to airplane transponders, which can be transmitted to ground stations using low-energy communication technologies like ADS-B. The UAV's 3D location (x, y, z) and speed will be tracked in real-time using GPS or GNSS, allowing ground stations to monitor its path. Sharing this data is essential for air traffic control and integrated systems to ensure safe distances between drones and people.

- **Collision Avoidance system**

To avoid collisions, UAVs should use computer vision and sensors like LiDAR, radar, and cameras to detect and steer clear of other drones. They can share location data through specific communication protocols for better coordination. Additionally, UAVs can integrate with existing air traffic control systems to automatically change course if a manned aircraft is nearby. AI-based algorithms can help adjust flight paths in real-time to prevent collisions..

- **Bird Deterrent system**

A Bird Deterrent System uses ground-based ultrasonic emitters placed near sensitive areas like airports or wildlife sanctuaries. These devices emit frequencies that safely warn birds to stay away from UAVs, preventing potential collisions without harming them.

SAFETY BACK-UP SOLUTIONS



- **Partial rotor failure**

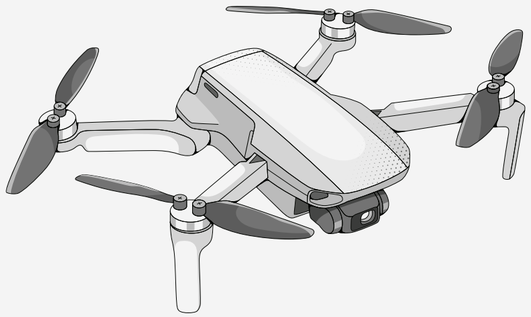
In the case of Hexacopter and Octacopter, the propulsion redundancy mechanism starts, where if a motor or two fail, the drone can safely maintain stability till it reaches the nearest **underground drone parking spot** for maintenance and for power supply.

- **Total rotor failure**

In addition, in the event of a total rotor failure, a **parachute deployment system (PDS)** is activated alongside with a dual purpose float, that performs as both a cushioning on land and a floatation device on water. For ensuring more safety, the use of a **flight termination system (FTS)**, that would force stop the propellers so that the parachute can operate properly, would be beneficial.

MITIGATING PRIVACY CONCERNS

- **Dynamic No-Fly Zones**: Local governments need a way to create temporary restricted airspace for events or hazards. UAV systems can connect to a centralized airspace management platform to get real-time updates and avoid these zones automatically.
- **Route Modification**: If a UAV nears restricted airspace, its onboard computer should quickly find a safe detour to avoid the no-fly zone. This can be done using UAV software with pre-installed, real-time updating geofencing.
- **Real-Time Monitoring and Editing**: Authorities can add or remove restricted areas based on current events, like establishing a temporary no-fly zone over a disaster site.
- **Passive Camera Detection system**, like RF or millimeter-wave, to check if UAVs have cameras, without interfering with their avionics.



NOISE REDUCTION TECHNOLOGIES



- **Propeller Design**

Structure

The silent Zipline Propeller with a serrated blade design, which is inspired by owl wings, breaks up the airflow over the blades; therefore, reducing turbulence and noise. While minimizing the turbulence, it’s optimized with efficient airflow and thrust by using a 30-degree blade angle. Furthermore, the asymmetrical blade shape works on reducing drag and improving lift besides the flexibility to absorb mechanical vibrations and even out the airflow, providing a quieter operation. The root of each blade has a wider chord, generating more lift at lower RPMs.

Material

Recycled carbon fibers & fiberglass filled with nylon would give the utmost efficiency as promised by several reputable companies such as ELSA industries

Propeller Design

Feature	Zipline Silent Propeller	Toroidal Propeller	Normal Propeller
Noise Level (dB)	**30-40 dB** (very low, near-silent)	**50-60 dB** (moderate reduction)	**70-80 dB** (standard drone noise)
Tip Vortex Reduction	**90-95%** (minimal vortex creation)	**80-85%** (significant reduction)	**40-50%** (high vortex generation)
Thrust Efficiency (%)	**85-90%** (optimized for low RPM)	**80-85%** (improved over standard)	**65-75%** (standard, drops at low RPM)
Drag Coefficient (Cd)	**0.2-0.25** (very low drag)	**0.3-0.35** (reduced drag, but higher than Zipline)	**0.5+** (higher drag due to flat blade design)
Material Weight (g)	**10-20 g** (light, advanced composites)	**15-25 g** (heavier due to loop design)	**10-15 g** (light, standard materials)
Manufacturing Cost ($)	**$200-300** (precision manufacturing)	**$100-150** (complex but less than Zipline)	**$10-30** (mass-produced, simple design)
Operating RPM Range	**500-3000 RPM** (low RPM for low-noise operation)	**1000-4000 RPM** (efficient at mid RPM)	**3000-8000 RPM** (best at high RPM, noisy)
Vortex Shedding	**Minimal** (advanced edge shaping)	**Low** (loop design controls shedding)	**High** (standard, uncontrolled)
Airfoil Profile	Custom, optimized for quiet airflow	Curved, reducing drag and vortices	Standard flat or curved blade



Normal Propeller



Toroidal

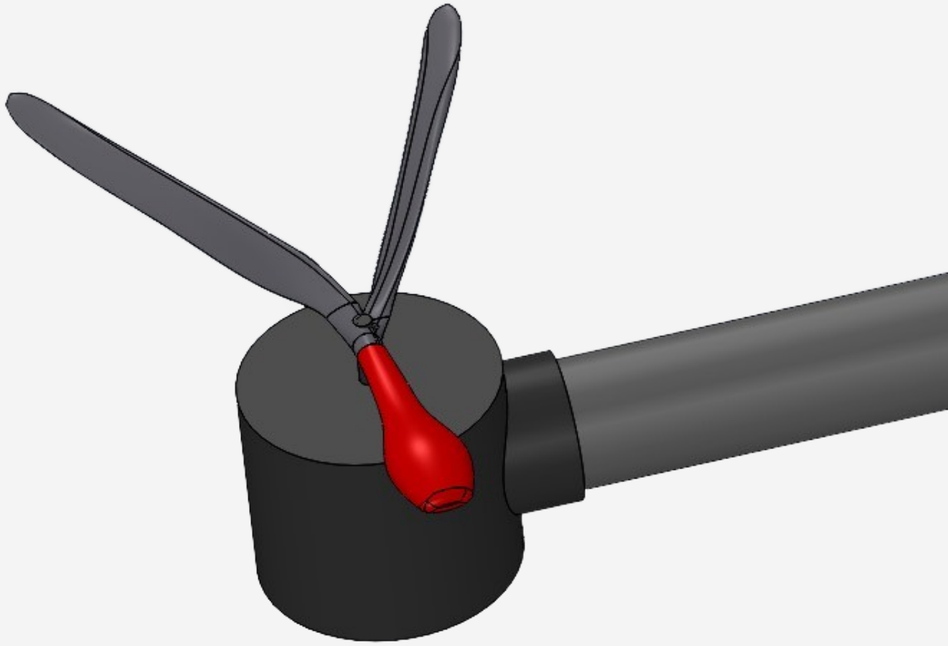


Zipline

NOISE REDUCTION TECHNOLOGIES

- **Noise dampening materials**

Polyurethane foam is an effective padding material for drones, as it absorbs noise from motors and propellers while being lightweight for optimal performance. Its durability and versatility allow for targeted noise reduction and vibration damping, enhancing stability. Affordable and easy to shape, it contributes to quieter flights and better acceptance in urban areas.



ENVIRONMENTAL IMPACTS SOLUTIONS



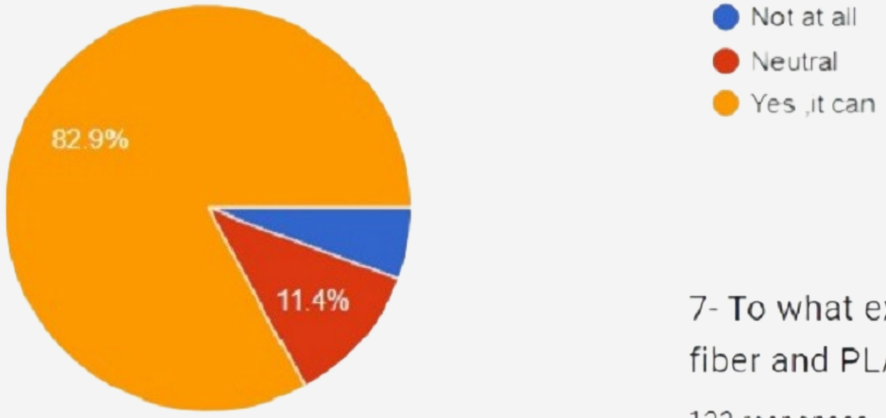
- **Power Management System Battery Efficiency** it is recommended to Implement an advanced power management system to extend battery life, which is crucial for long-distance flights or missions requiring high energy (e.g., operating sensors).
- Efficient energy distribution among sensors, propulsion, and communication systems will reduce the overall environmental impact.

- Reducing environmental impact and Enhancing Privacy using eco-friendly materials such as **recycled carbon fiber & PLA** for drone construction to lower the environmental impact. Additionally, ensure the propulsion system is as efficient as possible to reduce emissions (if any).

PUBLIC SURVEYS & FEEDBACK

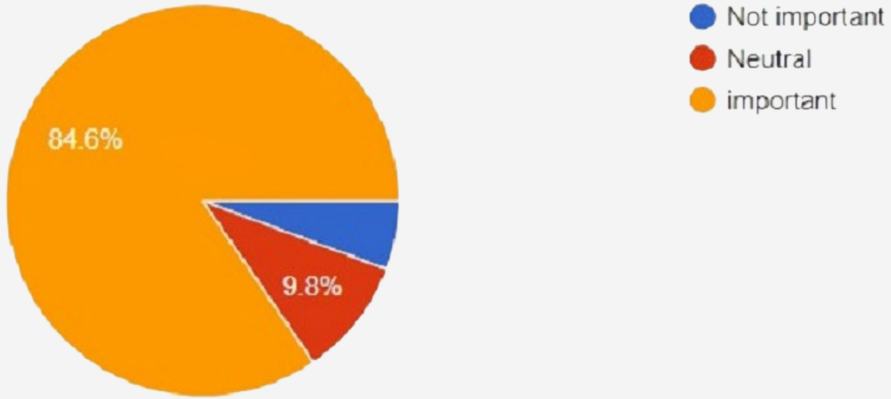
4-Do you think that implementing a radar system that detects UAVs with shooting cameras in No-photography zones (Residential areas) and then alerts the authorities would help in your acceptance of the UAVs?

123 responses



7- To what extent do you value the use of eco-friendly materials, like recycled carbon fiber and PLA, in UAVs construction to minimize environmental impact?

123 responses



The survey we made after posing the enhanced solutions to the issues at hand were positive so we expect the UAVs to be present in the upcoming future with almost no concerns and with utmost acceptance.

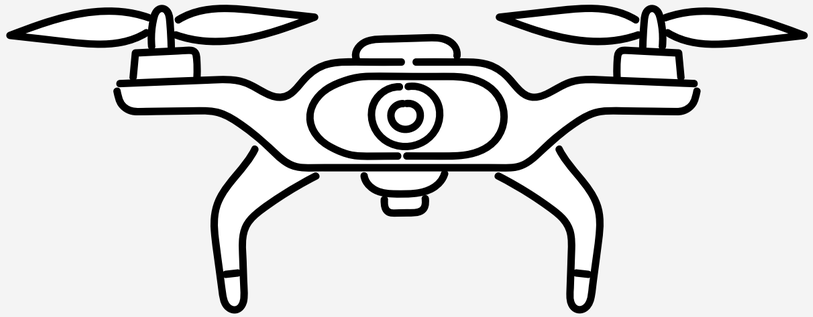
FINANCIAL Plan

Initial Hardware and Device Costs

Device	Cost per Unit (USD)	Quantity Needed	Total Cost (USD)
Remote ID Transmitter	$50	1 per UAV	$50
GPS/GNSS Receiver	$30	1 per UAV	$30
Communication Module (Wi-Fi/Cellular)	$40	1 per UAV	$40
Mesh Network Communication Module	$30	1 per UAV	$30
Encrypted Communication Module	$20	1 per UAV	$20
LiDAR Sensor	$300	1 per UAV	$300
Radar Sensor	$250	1 per UAV	$250
Optical Camera (for navigation)	$100	1 per UAV	$100
Onboard AI Processor	$150	1 per UAV	$150
Noise Damping Materials	$100	1 per UAV	$100
Ultrasonic Bird Deterrent	$50	1 per UAV	$50
Power Management System	$200	1 per UAV	$200
Total Cost			**$1,570**

Ground-Based Infrastructure Costs

Infrastructure/Device	Cost per Unit (USD)	Quantity	Total Cost (USD)
RF Scanner/Receiver (for camera detection)	$1,500	10 units	$15,000
Camera Detection System (passive RF)	$2,000	5 units	$10,000
No-Fly Zone Transmitter	$3,000	5 units	$15,000
Ground Control Stations	$50,000	2 stations	$100,000
Bird Deterrent System	$500	10 units	$5,000
Local Authority Management Terminals	$2,500	5 terminals	$12,500
Subtotal Ground-Based Costs			**$157,500**



Operational and Maintenance Costs

Item	Annual Cost (USD)	Notes
Software Maintenance & Updates	$150,000	Annual software updates, bug fixes
Ground-Based Equipment Maintenance	$25,000	Includes RF scanners, control stations
UAV Fleet Maintenance	$100,000	UAV hardware maintenance
Data Storage & Cloud Infrastructure	$75,000	Cloud-based data storage for UAV operations
Regulatory Compliance & Certifications	$50,000	Ensuring compliance with aviation authorities
Training Programs for UAV Operators	$40,000	Annual training and re-certifications

Software Development and Integration

Software Component	Development Cost (USD)	Notes
Centralized U-Space Management Platform	$300,000	Real-time tracking, flight planning, and control
Predictive Weather Analysis Integration	$50,000	Includes AI-based weather pattern prediction
Environmental Adaptation Algorithms	$40,000	Dynamic route planning based on environment
Swarm Intelligence for Collaborative UAVs	$200,000	AI-based swarm coordination and optimization
AI Collision Avoidance & Navigation System	$150,000	Integrated AI for navigation and avoidance
Geofencing & Dynamic Restricted Area Control	$100,000	Software for no-fly and no-photography zones
Real-Time Data Sync with Authorities	$50,000	Integration of live airspace updates
Public Engagement and Transparency Portal	$80,000	Portal for public transparency
User-Friendly Mobile App for Authorities	$50,000	App for managing restricted zones
Total Software Development Costs	**$1,020,**	


ANNOYING DRONE NOISES?

REFERENCES

The new EU regulatory framework for U-space. (n.d.). https://www.ecac-ceac.org/activities/unmanned-aircraft-systems/uas-bulletin/22-uas-bulletin/505-uas-bulletin-2-the-new-eu-regulatory-framework-for-u-space

Kim, T. (2016). Reduction of Tonal Propeller Noise by Means of Uneven Blade Spacing. https://escholarship.org/uc/item/9q75v9t9

What is U-space? (n.d.). https://www.ecac-ceac.org/activities/unmanned-aircraft-systems/uas-bulletin/22-uas-bulletin/504-uas-bulletin-2-what-is-u-space

What is U-space? (n.d.). https://www.ecac-ceac.org/activities/unmanned-aircraft-systems/uas-bulletin/22-uas-bulletin/504-uas-bulletin-2-what-is-u-space

Teja, H. S., Chawan, G., Nilay, S., Eswaraiah, D., & Jyothi, U. (2023). Design and analysis of drone propeller by using aluminium and nylon materials. E3S Web of Conferences, 391, 01032. https://doi.org/10.1051/e3sconf/202339101032

